

Sliding Window Xor

Time limit: 1.00 s

Memory limit: 512 MB

You are given an array of n integers. Your task is to calculate the bitwise xor of each window of k elements, from left to right.

In this problem the input data is large and it is created using a generator.

Input

The first line contains two integers n and k : the number of elements and the size of the window.

The next line contains four integers x , a , b and c : the input generator parameters. The input is generated as follows:

- $x_1 = x$
- $x_i = (ax_{i-1} + b) \bmod c$ for $i = 2, 3, \dots, n$

Output

Print the xor of all window xors.

Constraints

- $1 \leq k \leq n \leq 10^7$
- $0 \leq x, a, b \leq 10^9$
- $1 \leq c \leq 10^9$

Example

Input	Output
8 5 3 7 1 11	0

Explanation: The input array is $[3, 0, 1, 8, 2, 4, 7, 6]$. The windows are $[3, 0, 1, 8, 2]$, $[0, 1, 8, 2, 4]$, $[1, 8, 2, 4, 7]$ and $[8, 2, 4, 7, 6]$, and their xors are 8, 15, 8 and 15. Thus, the answer is $8 \oplus 15 \oplus 8 \oplus 15 = 0$.